Saransh Sharma

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Education

Rutgers University **Sep 2022 – May 2024**
Master of Science in Computer Science *CGPA: 3.8/4*

- **Teaching Assistant:** Mentored 250+ students in Data structures, algorithms and RDBMS

Manipal University Jaipur **Aug 2018 – Jul 2022**
Bachelor of Technology in Computer Science & Engineering *CGPA: 8.31/10*

Technical Skills

Languages & Backend: Java, Scala, Python, TypeScript, SQL, C++, FastAPI, Django, NodeJS, ReactJS, REST, GraphQL
Cloud & Data: AWS (Lambda, DynamoDB, S3, ECS, EventBridge, CDK, CloudWatch), Apache Spark
Tools, DBs & GenAI: Docker, Kubernetes, Git, CI/CD, PostgreSQL, MongoDB, LangChain, RAG, Vector DBs, Ollama

Experience

Amazon ***Software Development Engineer I*** **Oct 2025 – Present**
Social-ads org – publishes Amazon ads across Meta, Pinterest, TikTok, Snap & X

- Optimized a high-volume **ad-bidding** workflow, reducing compute & third-party **API spend by an estimated 20–30%**, by building an **event-driven data pipeline** (**Java, AWS Lambda, S3, SES, Scala/Spark**) across **3 AWS regions**
- Built a **reporting microservice** (Java, Lambda, S3, SES) that emails multi-region summaries, replacing manual reports
- Launched a **target return-on-ad-spend (tROAS)** bidding strategy, driving a **20% lift in ad returns**
- Migrated a production data pipeline off legacy **S3** object storage onto the org **data lake** (Scala/Spark), cutting ~**40 on-call tickets/month** and reducing data delays for downstream **ad-publishing** jobs
- Migrated a **5-stack production service** (VPC, Lambda, ECS, S3) to **IaC** (AWS CDK), reducing deployment risk

Incedo ***Software Engineer Intern*** **May 2025 – Sep 2025**

- Spearheaded design & development of a Retrieval-Augmented Generation **(RAG)** pipeline using **Ollama, ChromaDB, LangChain & HuggingFace APIs** to power **natural language Q&A** over structured Root Cause Analysis data
- Built a **SQL / knowledge agent workflow** to improve retrieval and answer accuracy over numerical data

Blue Sigma ***Software Engineer Intern*** **Sep 2024 – Dec 2024**

- Built a robust backend in **Python** & **Django**, exposing RESTful APIs for user authentication and data retrieval, and real-time data visualizations that **reduced page re-renders by 15%**
- Containerized deployments, ensuring consistency and **reducing deployment failures by 20%**

VigilanceAI LLC ***Software Engineer Intern*** **Jan 2022 – Apr 2022**

- Collaborated with engineers to design and deploy a **CNN-RNN model** for object, activity, and posture recognition, integrating it into a production-ready web application for real-time detection with **97% accuracy**
- Built a full-stack app (Node.js, **React.js**, **MongoDB**) with **RESTful APIs** linking the UI, backend, DB and ML models

Projects

Receipt Processor

- Built a RESTful API using **Python** and **FastAPI** to process JSON receipt data and compute reward points via rule-based logic, with modular design separating routing, validation, and business logic
- Enforced strict **input validation using Pydantic** with regex constraints and custom error handling; containerized with Docker, designed stateless/scalable, and tested with **Pytest** (**+60% coverage**)

HL7 Message Parser

- Engineered a Python **CLI parser** converting **healthcare** HL7 appointment messages into structured **JSON** via custom segment logic and Pydantic schema validation; Dockerized with structured logging and unit tests

Better, Smarter, Faster

- Simulated an Agent–Prey–Predator pursuit on a circular graph (probabilistic prey, greedy predator); implemented **Markov Chains** and a custom **Neural Network** to optimize the agent's decisions, **reducing time-to-catch by 5%**